

Calculus I Lesson 47B

Formula A: $\int \frac{1}{\sqrt{a^2 - u^2}} du = \arcsin\left(\frac{u}{a}\right) + C$

Formula B: $\int \frac{1}{a^2 + u^2} du = \frac{1}{a} \arctan\left(\frac{u}{a}\right) + C$

Formula C: $\int \frac{1}{u\sqrt{u^2 - a^2}} du = \frac{1}{a} \operatorname{arcsec}\left(\frac{|u|}{a}\right) + C$

Question 1:

$$\int \frac{1}{a^2 + u^2} du = \frac{1}{a} \arctan\left(\frac{u}{a}\right) + C$$

Find $\int \frac{1}{16 + x^2} dx$.

Hint: Let $a^2 = 16 \Rightarrow a = 4$

Let $u^2 = x^2 \Rightarrow u = x$

Hence, $\frac{du}{dx} = 1 \Rightarrow du = dx$

a) Write the integral $\int \frac{1}{16 + x^2} dx$ in terms of u : $\int \frac{1}{16 + x^2} dx = \underline{\hspace{2cm}}?$

b) $\int \frac{1}{16 + x^2} dx = \underline{\hspace{2cm}}?$

Question 2:

$$\int \frac{1}{a^2 + u^2} du = \frac{1}{a} \arctan\left(\frac{u}{a}\right) + C$$

Find $\int \frac{1}{16 + 9x^2} dx$.

Hint: Let $a^2 = 16 \Rightarrow a = 4$

Let $u^2 = 9x^2 \Rightarrow u = 3x$

Hence, $\frac{du}{dx} = 3 \Rightarrow du = 3dx$ and $\frac{1}{3} du = dx$

a) Write the integral $\int \frac{1}{16 + 9x^2} dx$ in terms of u : $\int \frac{1}{16 + 9x^2} dx = \underline{\hspace{2cm}}$

b) $\int \frac{1}{16 + 9x^2} dx = \underline{\hspace{2cm}}$

Question 3:

$$\int \frac{1}{a^2 + u^2} du = \frac{1}{a} \arctan\left(\frac{u}{a}\right) + C$$

Find $\int \frac{5}{25 + 36x^2} dx$.

Hint: Let $a^2 = 25 \Rightarrow a = 5$

Let $u^2 = 36x^2 \Rightarrow u = 6x$

Hence, $\frac{du}{dx} = 6 \Rightarrow du = 6dx$ and $\frac{1}{6} du = dx$

a) Write the integral $\int \frac{5}{25 + 36x^2} dx$ in terms of u : $\int \frac{5}{25 + 36x^2} dx = 5 \cdot \int \frac{1}{25 + 36x^2} dx = \underline{\hspace{2cm}}?$

b) $\int \frac{5}{25 + 36x^2} dx = 5 \cdot \int \frac{1}{25 + 36x^2} dx = \underline{\hspace{2cm}}?$

Question 4:

$$\int \frac{1}{a^2 + u^2} du = \frac{1}{a} \arctan\left(\frac{u}{a}\right) + C$$

Find $\int \frac{1}{8+11x^2} dx$.

Hint: Let $a^2 = 8 \Rightarrow a = \sqrt{8}$

Let $u^2 = 11x^2 \Rightarrow u = \sqrt{11} \cdot x$

Hence, $\frac{du}{dx} = \sqrt{11} \Rightarrow du = \sqrt{11} \cdot dx$ and $\frac{1}{\sqrt{11}} du = dx$

a) Write the integral $\int \frac{1}{8+11x^2} dx$ in terms of u : $\int \frac{1}{8+11x^2} dx = \underline{\hspace{2cm}}$

b) $\int \frac{1}{8+11x^2} dx = \underline{\hspace{2cm}}$

Question 5:

$$\int \frac{1}{a^2 + u^2} du = \frac{1}{a} \arctan\left(\frac{u}{a}\right) + C$$

Find $\int \frac{12}{7+13x^2} dx$.

Hint: Let $a^2 = 7 \Rightarrow a = \sqrt{7}$

Let $u^2 = 13x^2 \Rightarrow u = \sqrt{13} \cdot x$

Hence, $\frac{du}{dx} = \sqrt{13} \Rightarrow du = \sqrt{13} \cdot dx$ and $\frac{1}{\sqrt{13}} du = dx$

a) Write the integral $\int \frac{12}{7+13x^2} dx$ in terms of u : $\int \frac{12}{7+13x^2} dx = 12 \cdot \int \frac{1}{7+13x^2} dx = \underline{\hspace{2cm}}$

b) $\int \frac{12}{7+13x^2} dx = 12 \cdot \int \frac{1}{7+13x^2} dx = \underline{\hspace{2cm}}$

